

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.4: Independence, bases, and rank

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Linear independence

Activity

In the activity we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In the activity, we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W . The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W . But why is S ‘better’?

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the ‘most efficient’. In data terms, independence asks whether one feature direction is already built from the others.

Definition: Linear Independence A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be

linearly independent

, or simply

independent

if the only solution to the equation $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$ is trivial, i.e., $t_1 = \dots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is

linearly dependent

if it is *not* linearly independent, i.e.,

if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

The last example demonstrates an algorithm to verify whether a given set of vectors is linearly independent.

To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.
2. Solve the resulting linear homogeneous system.
3. If there is a nontrivial solution to the equation, then the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M . Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ has a unique representation as a linear combination of the $\mathbf{x}_i$.

Proof Let $\mathbf{x}$ be an element of $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \dots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \dots + b_k\mathbf{x}_k,$$

then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \dots + (a_k - b_k)\mathbf{x}_k = \mathbf{0}.$$

Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1 \mathbf{x}_1 + \dots + t_k \mathbf{x}_k = 0.$$

But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

Activity

Is the set $S = \{\mathbf{0}\}$ linearly independent?

Does the equation $t\mathbf{0} = \mathbf{0}$ have a nontrivial solution?

Fact The nonzero rows of a matrix in row echelon form are linearly independent.

Basis and dimension

In geometry, we often describe lines as being ‘1-dimensional’, and planes as being ‘2-dimensional’. We can describe the notion of dimension more precisely using linear algebra.

Theorem: A Fundamental Theorem Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

Definition: Bases of Vector Spaces If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ in U is called a **basis** of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.

Theorem: The Invariance Theorem If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

The Invariance Theorem says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition: Dimensions of Vector Spaces The **dimension** of

a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U . We often write $\dim(U)$ for the dimension of U , and if $\dim(U) = d$, we say U is d -dimensional. By convention, the dimension of the trivial subspace $\{0\}$ is *zero*.

- **Term:** spanning set
Meaning: enough directions to generate the whole subspace
- **Term:** independent set
Meaning: no direction is redundant
- **Term:** basis
Meaning: just enough directions: spanning and independent
- **Term:** dimension
Meaning: number of directions in any basis

Activity

What is the dimension of $\mathbb{R}^3$?

Activity

What is the dimension of $\mathbb{R}^4$?

Activity

What is the dimension of $\mathbb{R}^n$?

Activity

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

Fact Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U .
- Any spanning set for U contains a subset of vectors which form a basis for U .

Theorem Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then B is linearly independent if and only if B spans U .

Proof Suppose B is linearly independent and $|B| = m = \dim(U)$. By A Fundamental Theorem, one cannot add any new vectors to the set B and remain linearly independent. But the referenced item says that if B is not already a spanning set for U , one can add new vectors to B from U , and remain linearly independent. Thus B must be a basis.

Conversely, if B spans U , then by the referenced item we can choose a subset of B which is a basis for U . But any proper subset of B contains fewer than m vectors, and thus by The Invariance Theorem, cannot be a basis for U . Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v}, \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel, i.e., if one vector is a scalar multiple of the other.

Activity

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$. See textbook figure fig-5-2-independent and textbook figure fig-5-2-dependent.

Note More generally, if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Activity

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}.$$

Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination of a vector in the subspace are the coordinates.

Definition: Ordered Bases of a Subspace Let $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a ba-

sis of the subspace U . We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an

ordered basis

of U .
As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$, and so describe the same bases of $\mathbb{R}^2$.

Note Many Linear Algebra textbooks, including *Linear Algebra With Applications*, *Nicholson* write ordered bases using set notation $\{\dots\}$, despite the order in which the basis elements appear being important. We use tuple notation $(\dots)$ to emphasize the importance of the order.

Given an ordered basis B for an m -dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its ‘coordinates’ with respect to the basis B , identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition: Coordinate Vectors With Respect to a Basis Let U be an m -dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U . Given any vector $\mathbf{v} \in U$, there exists unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1 \mathbf{b}_1 + \dots + t_m \mathbf{b}_m$$

and we define the

coordinate vector

of a vector $\mathbf{v} \in U$ with respect to B to be the m -vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}.$$

See textbook figure fig-5-2-coordinate-basis-diagram for an illustration of this concept, taken from *Stanford’s MATH 51 Textbook*.

Note If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U , then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector in $\mathbb{R}^m$.

Activity

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}.$$

Rank of a matrix

We are now going to connect the concept of dimension with the notion of rank. In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined analogously as how they are defined for column vectors.

Definition: Column and Row Space Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$ span a *subspace* of $\mathbb{R}^m$ called the **column space** of A , denoted $\text{col}(A)$.
- The rows of A , considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the **row space** of A , denoted $\text{row}(A)$.

Note The column space is the image space from Subspaces written with a column-based name:

$$\text{col}(A) = \text{im}(A).$$

The notation $\text{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\text{col}(A)$ emphasizes the columns that span those outputs.

Lemma Let A and B be two row equivalent $m \times n$ matrices. Then $\text{row}(A) = \text{row}(B)$.

Proof Each time we apply a row operation to a matrix, we replace one or more of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A . But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A' . Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Lemma If R is a matrix in row echelon form (recall Row echelon form and reduced row echelon form) then the non-zero rows of R are a basis of $\text{row}(R)$.

Proof The nonzero rows of R are linearly independent by the referenced item, and span $\text{row}(R)$ by definition. They thus by definition form

a basis for $\text{row}(R)$.

Remark In Rank of a matrix, we defined the *rank* of a matrix as the number of leading 1s (i.e., the number of pivots) of any matrix R in row echelon form which is row-equivalent to A . By the referenced item, $\text{rank}(A) = \dim(\text{row}(A))$, which justifies that the rank of A is independent of the choice of matrix R .

Activity

Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Theorem: Rank Theorem Let A denote any $m \times n$ matrix of rank r . Then

$$\dim(\text{col}(A)) = \dim(\text{row}(A)) = r.$$

Moreover, if A is row-equivalent to a matrix R in row echelon form, then

- The r leading rows of R are a basis for $\text{row}(A)$.
- If the pivots of R lie in columns $j_1, j_2, \dots, j_r$, then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\text{col}(A)$.

Warning: Pivot columns come from the original matrix Row reduction identifies pivot positions, but row operations do not preserve

the original columns of A . When finding a basis for $\text{col}(A)$, use the pivot columns of the original matrix A , not the columns of the row-reduced matrix.

- **Goal:** basis for $\text{row}(A)$
Row reduction tells you: nonzero rows of REF/RREF
Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\text{col}(A)$
Row reduction tells you: pivot column positions
Use these vectors: corresponding columns of original A
- **Goal:** basis for $\text{null}(A)$
Row reduction tells you: free variables
Use these vectors: basic solution vectors in $\mathbb{R}^n$
- **Goal:** rank
Row reduction tells you: number of pivots
Use these vectors: a number
- **Goal:** nullity
Row reduction tells you: number of free variables
Use these vectors: a number

Fact It follows from Rank Theorem that for any matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
What is $\dim(\text{row}(A))$?

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the column space of A .

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
What is $\dim(\text{col}(A))$?

Activity

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to the activity.
Find a basis for the nullspace of A .

Definition The nullity of a matrix A is equal to $\dim(\text{null}(A))$. It is denoted $\text{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The row-reduction definition of rank now connects to dimension.

Theorem: Rank-Nullity Theorem Let A denote an $m \times n$ matrix of rank r . Then

- The $n - r$ basic solutions to the equation $Ax = 0$ provided by the Gaussian algorithm are a basis for $\text{null}(A)$, so $\dim(\text{null}(A)) = n - r$.
- $\dim(\text{im}(A)) = r$.
- $\dim(\text{im}(A)) + \dim(\text{null}(A)) = n$.
- $\text{rank}(A) + \text{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \text{rank}(A) + \text{nullity}(A)$$

We can now state one of the preview patterns from Section 2.1, introduced in Two inputs, one output, in the language of null spaces.

Activity: Same output from a null-space direction

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness?

Activity: Revisiting the height map with rank and nullity

Return again to A map that forgets height. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P . How does rank-nullity describe the height direction that P forgets?

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{im}(B)$.

Activity

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{null}(B)$.

Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.

Rows below represent houses. The three features are

first-level area, second-level area, total area,

measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X . The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

1. Compute $X\mathbf{z}$.
2. What does this say about the three feature columns?
3. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Compute $X\mathbf{c}$ and $X\mathbf{c}'$.

4. The two coefficient vectors tell different stories. Why do they make the same predictions?
5. What warning does this give about asking which feature is most important?

Note: Common exact feature redundancies Common exact feature redundancies include:

- first-level area, second-level area, and total area;
- minutes and seconds, where $\text{seconds} = 60 \cdot \text{minutes}$;
- height in inches and height in centimeters, where $\text{centimeters} = 2.54 \cdot \text{inches}$;
- subtotal, tax, tip, and total bill, where $\text{total} = \text{subtotal} + \text{tax} + \text{tip}$;
- exam parts and exam total, where $\text{total} = \text{part 1} + \text{part 2} + \text{part 3}$.

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Which features are linearly redundant?

For each feature list, decide whether one feature is a linear combination of the others.

1. first-level area, second-level area, total area;
2. height in inches, height in centimeters;
3. width, height, area of a rectangle;
4. quiz 1 score, quiz 2 score, quiz total;
5. bedrooms, bathrooms, selling price.

Activity: Auditing redundant features in code

The matrix below stores the house features from the previous activity.

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector c' .

Output:

```
(array([0, 0, 0, 0]), 2, array([34, 42, 42, 32]), array([34, 42, 42, 32]))
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?
3. Why do Xc and Xc' agree?
4. What does this say about interpreting coefficients when features are redundant?

Activity: A difference matrix forgets level

Let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

Activity: Reading the difference matrix in code

```
D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])

x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
(array([3, 4, 1]), array([3, 4, 1]), array([0, 0, 0]))
```

1. What does $D @ x$ measure?
2. Why do the first two outputs agree?
3. What does $D @ \text{ones}$ say about the map?

Redundant features create null-space directions. If $Xz = 0$, then

$$X(\mathbf{c} + t\mathbf{z}) = X\mathbf{c}$$

for every scalar t . Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.